

ex2.A114290027

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Question 1

```
library(ggplot2)

commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38,
            29, 24, 31, 36, 40, 26, 33, 28, 21, 44,
            30, 37, 23, 39)

a <- cut(commute, 5)

freq <- table(a)

cum_freq <- cumsum(freq)

upper_boundaries <- c(23.4, 28.8, 34.2, 39.6, 45)

ogive_data <- data.frame(
  upper_boundary = upper_boundaries,
  cumulative_frequency = cum_freq
)

freq
```

```
## a
## (18,23.4] (23.4,28.8] (28.8,34.2] (34.2,39.6] (39.6,45]
##          5          5          5          5          4
```

```
cum_freq
```

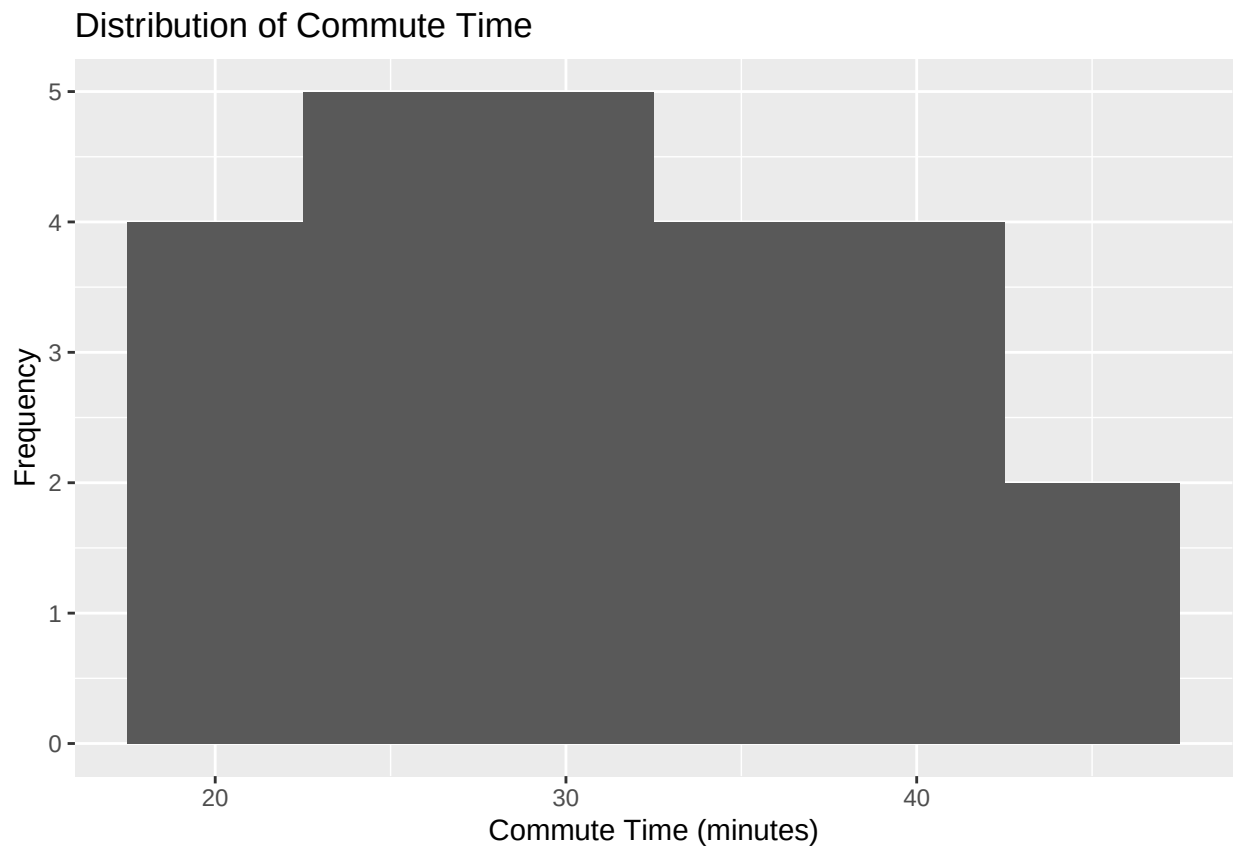
```
## (18,23.4] (23.4,28.8] (28.8,34.2] (34.2,39.6] (39.6,45]
##          5          10          15          20          24
```

```
ogive_data
```

```
##          upper_boundary cumulative_frequency
## (18,23.4]          23.4              5
## (23.4,28.8]         28.8              10
## (28.8,34.2]         34.2              15
## (34.2,39.6]         39.6              20
## (39.6,45]          45.0              24
```

Question 2

```
ggplot(data.frame(commute), aes(x = commute)) +  
  geom_histogram(binwidth = 5) +  
  labs(  
    title = "Distribution of Commute Time",  
    x = "Commute Time (minutes)",  
    y = "Frequency"  
  )
```



The distribution is approximately symmetric.

Question 3

```
ggplot(ogive_data, aes(x = upper_boundary,  
                        y = cumulative_frequency)) +  
  geom_line() +  
  geom_point() +  
  labs(  
    title = "Ogive of Commute Time",  
    x = "Upper Boundary",  
    y = "Cumulative Frequency"  
  )
```

